Is the U.S. Government spending its $350 million in wildlife corridor funding wisely?

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Overview

Intended Audience: U.S. Department of Transportation, State Government Officials and State Transportation Agencies

The primary cause of biodiversity loss is habitat loss and habitat fragmentation. Habitat fragmentation happens when parts of a habitat are destroyed, leaving behind smaller unconnected areas. In the United States, habitats are often fragmented to build roads and freeways for human use. As a result, animals struggle to survive and can be the cause of motor vehicle accidents.

There are more than a million automobile accidents per year involving wildlife in the United States[2]. The amount spent on medical costs and vehicle repairs amount to more than $8 billion per year. In 2021, the U.S. Department of Transportation announced that it was allocating 350 million dollars to fund the construction of wildlife corridors[6]. Currently, the grant money has yet to be distributed.

The dwindling animal populations and animal-motor vehicle collision rate has become such a big problem that U.S. states have begun building wildlife corridors. Wildlife corridors are narrow strips of land that serve as traveling avenues for wildlife species between fragmented habitats[3]. However, data around existing corridors is scattered, and the features that make up an environmentally "successful" corridor are still unknown in most cases, which makes analysis difficult.

Existing Literature

In the past, each wildlife corridor project has been built on scattered data, none of which has been thoroughly analyzed enough to determine what features are necessary to build a corridor that will be used by the intended species populations, this is what we define as a "successful" corridor in this paper[7].

Existing literature does suggest that factors such as (1) reduced speed limits, (2) seasonal road closures, (3) raised road beds, (4) and crossing structures can increase the success rate of wildlife corridors[4,5]. There are knowledge gaps identified by experts in the field which include, but aren't limited to:

1. What are the critical dimensions a corridor needs?
2. Is there a critical threshold in intensity of land use?
3. How do landscape traits of corridors affect different species' likelihood to use them?

4. How far away from the linkage edge does human activity need to be managed?
5. How do time lags affect corridor use?
6. Which Governance and Management Activities Are Most Acceptable to People Living in or near Corridors and Which Policies Are Beneficial for Corridors?

Each of these remaining questions require studies to answer them. Given the large scope of work required to answer any of the above questions, our work will focus on one current knowledge gap. How does the length and width of a wildlife corridor impact the rate of species crossing by species type (i.e. predator species, prey species)?

Anticipated Impact

The goal of this study is to analyze two prominent structural features: length and width of wildlife corridors to assess the impact on the rate of species crossing the corridor. The effect of this explanatory research could be measured beyond the survival of multiple species due to the increase of food sources, life expectancy, genetic diversity, migration, and restoration of fragmented populations.[9] Wildlife corridors have the potential to create safer environments and communities for both animals and humans. In addition, they can also create jobs, promote driver safety, reduce the number of wildlife related automobile accidents and associated costs, and promote ecological restoration.

Research Question

The main focus of this study is to answer the following:

- How does the length and width of a wildlife corridor impact the rate of species crossing by species type?

We are also testing the following sub-questions:

- Is the relationship between length and width of a wildlife corridor the same for predator and prey species?
- What is the relationship between increasing the corridor area and the average size of the animals crossing?

Study Design

This explanatory study will conduct a mixed method research design. This approach will enable the collection of data from different sources such as publicly available data, surveys, interviews, and observations. The need for this type of design stems from the non-comprehensive accessible data that lacks details pertaining to the length and width features that comprise efficient wildlife corridors in relation to species. As a result, we will be conducting surveys, and phone interviews of employees that manage wildlife corridors to gain information about the aforementioned data points in each corridor. In addition, we will set up monitoring to calculate the proportion of animal crossing. This will provide this research with information about important factors such as population crossing, species types, and regional animal population sizes in the surrounding area. Controlling and keeping variables constant across wilderness areas is challenging so effort will be taken to normalize across different environments and seasons.

Once all data is gathered, it will be integrated to proceed with analysis. We plan to use all of the collected data to establish a baseline for the average rate of animal crossing, based on the specific habitat type. Once the baseline for a “successful” wildlife corridor has been established, we will add it to the dataset as a new column. We will add two columns to the table for tracking these divisions in successful vs. unsuccessful wildlife crossings. One column will contain the successful or unsuccessful corridor label for predator species, and another column will contain the successful or unsuccessful category for prey species. We are grouping each species into predator and prey because corridors are often built to accommodate one species more than others, so this will allow us to provide a more useful analysis.

After data has been finalized, we plan to use two different models: one that tests the significance of the relationship between predator species and the length and width of wildlife corridors. And another model that tests the significance of the relationship between prey species and the length and width of wildlife corridors.

Data

The study will collect data to analyze the impact of the length and width of corridors on the rate of species crossing the corridor.

- We will begin by compiling a list of all wildlife corridors within the sampling frame and determining if there is sufficient publicly available data on each corridor
- If we cannot find sufficient publicly available data on the corridor, we will reach out to all corridor managers on the list (as described in the study design).
- We will provide each manager by email
    - A survey asking about:
        - Corridor they maintain
        - Corridor habitat type
        - Length of corridor
        - Width of corridor
        - If the corridor was designed for a specific animal. And if yes, which animal?
        - List of predator and prey species in the area
        - List of predator and prey species that use the corridor
        - Number of animals crossing, broken down by species (if the information is available)
        - Population estimates for the prey and predator species previously provided
    - Permission to implement animal monitoring equipment on-site at their corridors (if none already implemented)

        - Animal trap (cameras) to count number of animals of each species crossing
        - This can also be used to set up a time-series study for later investigation
    - If we do not receive a response to the survey after 1 week, we will send a follow up email. If we do not receive a survey response within the following week, we will contact the manager of the corridor by phone, ask them the survey questions and record their responses.

Variables

Since we are creating an explanatory model, there is no experiment or intervention. We will be using the calculated column, crossing rate, to determine if a corridor is successful or not, which will provide the baseline for our model.

Crossing Rate = (total number of a species type)/ (total number of species type in the habitats connected by the corridor)

Corridors we deem as "successful", will have a crossing rate equal to or above the average for each species type (predator or prey). We will then use that information to calculate the average crossing rate of each species type, to get an average crossing rate for predator species, and an average crossing rate for prey species. Crossing rate is the most appropriate way to measure the success of a wildlife corridor, because it is one of the standard measures for corridor success in this industry[12].

We plan to control for externalities in this experiment by expanding the timeline from one year to three years, to account for changes such as breeding seasons, hibernation, and other factors.

Hypothesis

The motivation behind this observational study is to explain the statistical significance of the effect that length and width of a wildlife corridor has on the rate of animal crossings. We believe that this impact will vary depending on the type of species (i.e. whether the animal is a prey or predator species). In order to show this relationship, we will employ a two-phase explanatory sequential design with the procedure of connecting quantitative and qualitative data gathered from existing public databases and observations.

The quantitative data will come from the results of testing the following quantitative hypotheses:

- Null Hypothesis (H_0): The length and width of a wildlife corridor has no statistically significant impact on the rate of species crossing.
- Alternative Hypothesis (H_1): The length and width of a wildlife corridor has a statistically significant impact on the rate of species crossing.
    - Hypothesis for prey species model:
        - The length of the corridor has a negative relationship with the number of prey animals crossing it.

        - The width of the corridor has a positive relationship with the number of prey animals crossing it.
    - Hypothesis for predator species model:
        - The length of the corridor has a positive relationship with the number of predator animals crossing it.
        - The width of the corridor has a negative relationship with the number of predator animals crossing it.

The qualitative data will come from answering questions about corridor design, species, animal descriptions, predator-prey relationships, etc.

Sample

We will be using non-probability sampling for this project. The focus of our sample is countries that we believe can provide the most information, which is countries with a high number of wildlife corridors. Our sampling frame will include the following countries: Netherlands, Canada, Brazil, United States. Corridors outside the sampling frame will be excluded from analysis, but all corridors within the sampling frame countries will be included. There are over 1,200 corridors between these countries that we plan to collect data from[11].

Statistical Methods

We will use conventional statistical methods to obtain descriptive statistics such as mean, standard deviation, etc. to perform exploratory data analysis (EDA) of our quantitative data. In order to test our hypotheses, we will be running a linear regression for each model described in the study design (i.e. length, width on prey and predator species). We will use the t-test statistics to identify the causal relationship of length, and width dimensions on the success of a wildlife corridor between our study sample and the target population.

In order to analyze our qualitative data, we will apply a one-way ANOVA (Analysis of Variance) to examine how the variation between different species impacts the crossing rate for a specific corridor[1].

In addition, we will employ the Analytic Network Process (ANP) method to measure the weight of the structural features of length and width, and the species variety control to assess their critical roles in the successful design of a wildlife corridor. ANP will allow us to apply quantitative and qualitative attributes concurrently.

Potential Risks

Avoiding confounding variables: Wildlife corridors (or any wilderness setting) are inherently difficult to compare with each other because variables can be challenging to manage due to the dynamic environment.[10] Not all wildlife corridors will be from the same climate as each other, different species will be present at each corridor, and other external factors will differ across text studies.

We will do our best to mitigate this by holding variables in proportion to their environments. For example, when considering the number of predators crossing a specific corridor we will analyze that number in the context of the total population of predators that have access to said corridor.

The other big factor that differs is climate so we will have to normalize comparison across corridors by taking climate into account. That is, northern temperate winters should not be compared apples-for-apples with tropical climate winters.

Geographical Bias: We selected our countries due to closer geographies and easier time zones. The Netherlands is the exception and was included because it has one of the largest corridors in the world (right now). These geographies have some climates that vary, however, it is possible that the findings of our study are only suitable to be applied to similar wilderness areas and climates as those ultimately selected by the research team.[8] We will manage this by explicitly calling this out in our final research paper and all presentations throughout the project timeline.

Institutional Animal Care and Use Committee (IACUC) approval: Animals will be analyzed in this study so we will need to file with and receive approval for our study by the IACUC before we commence our research. We don't anticipate having a problem with this step.

State-by-state video laws: Some of the animal trap cameras will be in areas that humans may cross in front of and each US state or other country in our study has different consent laws for recording others. We will research these laws ahead of reaching out to the wildlife corridor managers so that our cameras are compliant with local regulations.

Additionally, we may wish for some camera traps to be located on private properties. If this is the case we will make sure to receive permission before installation. We will not install cameras if we are not provided permission.

Deliverables

Phase 1 [Two Months]: Identify all the wildlife corridors we would like in the study
- Conduct desktop research on existing wildlife corridors
- Collate a list of corridors to include in study and reasoning for inclusion
- Collect contact information of the wilderness corridors
- File study design with the Institutional Animal Care and Use Committee to receive approval

Phase 2 [Two Months]: Initial Contact
- Reach out to all corridors intended for inclusion in study
- Receive word and permission from participants

Phase 3.a [One Month]: Survey
- Send out survey to all participants

- Follow-up two weeks after initial survey contact to participants who have not responded

Phase 3.b [Three Years]: Camera Traps and Explanatory Study

- We are taking three years of data collection to account for changing weather conditions due to the seasons. We would like to avoid something like an unusually dry winter at one of the corridors skewing results. This is why we are taking three years worth of data.
- We will have cameras installed by the first month of this phase so that we can collect the average animal crossing rate per species at each corridor.
- Interim presentations will be given each year of the study

Phase 4 [Two Months]: Modeling

- This phase is where we run our statistical analysis on the collected data and test our hypotheses.

Phase 5 [Two Months]: Final Report

- Finalize report with lessons learned and recommendations
- Submit report to peer reviewed journal
- Give a final presentation to the Department of Transportation

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